

Abdelrahman Khalifa

Embedded Software Engineer — Zephyr · Embedded Linux · Rust

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SUMMARY

Embedded software engineer with over a year of professional firmware work on Zephyr RTOS and FreeRTOS, embedded Linux experience with Yocto and Buildroot, and upstream contributions to Zephyr. Writes C, modern C++ and Rust; completed Google Summer of Code 2026 with The Linux Foundation.

EXPERIENCE

Embedded Systems Engineer — EKSON | Aug 2025 – Present

- Developed Zephyr RTOS firmware for an ESP32-S3 3-DoF motion simulator, migrating it from FreeRTOS for more deterministic behavior and production readiness.
- Integrated an MPU6050 IMU over I2C with calibration, filtering and compensation for stable motion estimation.
- Built a modern C++/Qt desktop control application with reliable UART control of the simulator and optimized its performance by ~40% through asynchronous processing and fewer threads.

Google Summer of Code 2026 Contributor — The Linux Foundation / OpenPrinting | May 2026 – Oct 2026

COSMIC desktop printer setup tool in Rust | github.com/Abd002/GSOC-2026 | github.com/Abd002/cosmic-printers

- Built a Linux printer-management stack as 5 Rust crates with an async client, a CUPS/IPP/DNS-SD backend behind a varlink IPC service, and a shared UI; tested on real HP printers.
- Wrote FFI bindings from Rust to C (libcups3 DNS-SD API) and fixed crashes in libcups' Avahi code; **7 PRs merged upstream**.

Embedded Linux Systems Intern — ODC | Dec 2023 – Jan 2024

- Built Linux systems with Yocto and Buildroot; worked with kernel modules, device drivers, device trees, file systems and hardware flashing.

OPEN SOURCE

- **Zephyr RTOS** (Mar 2024): implemented the IP_MULTICAST_LOOP socket option for IPv4 across 6 networking modules (150+ lines), reviewed and merged upstream ([commit b117036](https://github.com/Abd002/Zephyr/pull/117036)).
- **OpenPrinting libcups and cups-rs** (2026): fixed DNS-SD crashes in libcups (C); ported cups-rs to libcups3 with CI ([PRs #18–#22](https://github.com/Abd002/cups-rs/pull/18)); Rust lp/lpstat in Winter of Code 2026.
- **Rust ecosystem** (Dec 2025): Boa JavaScript engine string optimization ([PR #4553](https://github.com/boa-dev/boa/pull/4553)) and a Rust Clippy lint fix ([PR #16214](https://github.com/rust-lang/rust-clippy/pull/16214)).

PROJECTS

- **AUTOSAR-Based V2X Ecosystem for SDV** (graduation project; FreeRTOS, AUTOSAR): software-defined vehicle prototype with ADAS (LKAS, driver monitoring), FOTA updates, and a secure bootloader with rollback. [GitHub](https://github.com/Abd002/AUTOSAR-V2X)
- **Seat Heater Control System** (Tiva C, FreeRTOS): real-time controller with temperature monitoring, heater-level control and UART diagnostics, using tasks, mutexes and event flags. [GitHub](https://github.com/Abd002/Seat-Heater-Control-System)
- **IoT Device Communication System** (C++, Raspberry Pi 4, QEMU, Yocto): TCP unicast and UDP multicast framework between a Pi server and emulated clients; Yocto cross-compilation and Make. [GitHub](https://github.com/Abd002/IoT-Device-Communication-System)
- **AUTOSAR PORT and GPIO drivers** (C, TM4C): developed and validated MCAL drivers following the AUTOSAR specification.
- **Traffic Light Controller** (Verilog, Spartan-3E FPGA): FSM with debounce and delay modules, cutting signal timing errors by 20%. [GitHub](https://github.com/Abd002/Traffic-Light-Controller)
- **Door Locker Security System** (C, two MCUs): password lock with keypad, LCD, I2C EEPROM, timer and DC motor; UART between the MCUs. [GitHub](https://github.com/Abd002/Door-Locker-Security-System)

TECHNICAL SKILLS

Languages: C, C++ (modern C++, Qt), Rust, Python, Bash, Verilog/VHDL

RTOS & firmware: Zephyr, FreeRTOS, ARM Cortex-M (STM32, TM4C), ESP32, AVR, AUTOSAR, MISRA C, bootloaders, FOTA

Embedded Linux: Linux (Ubuntu/Debian), operating systems, Yocto, Buildroot, kernel modules, device drivers, device trees, QEMU, Raspberry Pi

Protocols & tools: UART, SPI, I2C, CAN, LIN, USB, Ethernet, TCP/UDP; Git, CMake, Make, CI, testing, debugging, performance optimization, code review

EDUCATION

B.Sc. Computer and Systems Engineering — Minia University | 2020 – 2025

Very Good (84.6%) with Honors. Codeforces Expert and 4-time ECPC finalist; Deloitte Software Engineering Mentorship (2024).